

Fireflies with a Triangle-Wave Response

Complete solution to Exercise 4.5.1

We are given

$$\dot{\Theta} = \Omega, \quad \dot{\theta} = \omega + Af(\Theta - \theta),$$

where $A > 0$, and f is the 2π -periodic triangle wave

$$f(\varphi) = \begin{cases} \varphi & -\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2} \\ \pi - \varphi & \frac{\pi}{2} \leq \varphi \leq \frac{3\pi}{2} \end{cases}$$

Define the phase difference and frequency mismatch by

$$\varphi = \Theta - \theta, \quad \delta = \Omega - \omega.$$

Subtracting the two phase equations gives the single autonomous equation

$$\dot{\varphi} = \delta - Af(\varphi).$$

Everything below follows from this phase-difference equation.

(a) Graph of $f(\varphi)$

On $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the graph is the line $f(\varphi) = \varphi$, with slope $+1$. On $[\frac{\pi}{2}, \frac{3\pi}{2}]$, it is the line $f(\varphi) = \pi - \varphi$, with slope -1 . Periodic repetition produces a triangle wave of period 2π .

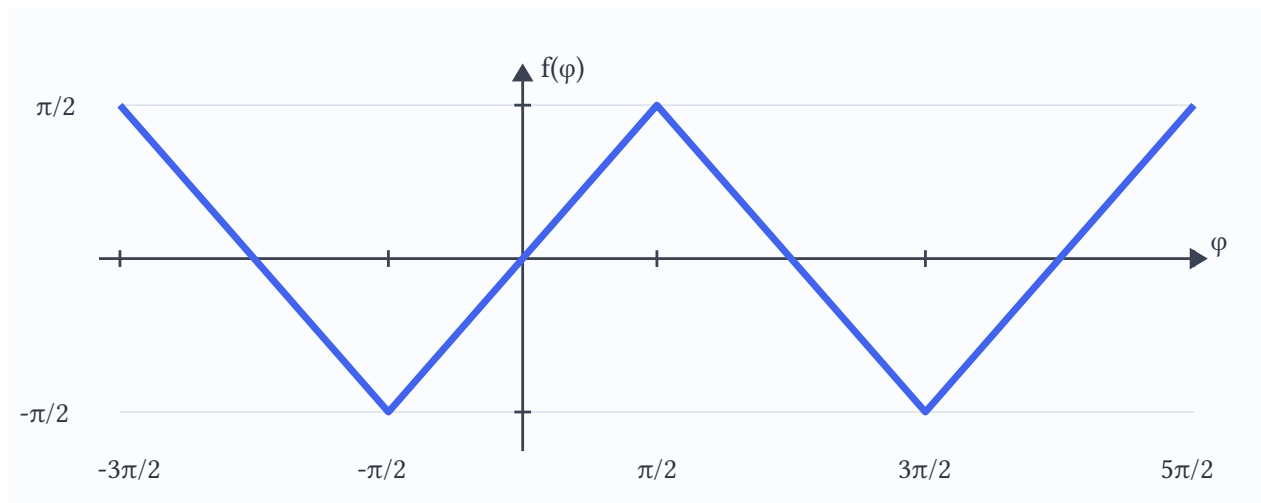


Figure 1: The 2π -periodic triangle wave. Its maximum is $\frac{\pi}{2}$, and its minimum is $-\frac{\pi}{2}$.

$$-\frac{\pi}{2} \leq f(\varphi) \leq \frac{\pi}{2}, \quad \text{period} = 2\pi.$$

(b) Range of entrainment

Entrainment means that the phases are locked: φ is constant. Therefore $\dot{\varphi} = 0$, so

$$0 = \delta - Af(\varphi^*) \rightarrow f(\varphi^*) = \frac{\delta}{A}.$$

Such a fixed point exists exactly when $\frac{\delta}{A}$ lies in the range of f . Since the range is $[-\frac{\pi}{2}, \frac{\pi}{2}]$,

$$-\frac{\pi}{2} \leq \frac{\delta}{A} \leq \frac{\pi}{2}.$$

Using $\delta = \Omega - \omega$, the entrainment condition is

$$|\Omega - \omega| \leq A\frac{\pi}{2}, \text{ or equivalently } \omega - A\frac{\pi}{2} \leq \Omega \leq \omega + A\frac{\pi}{2}.$$

At equality, the fixed point sits at a corner of the triangle wave. Just outside this interval, the phase slips continuously; the drift period found in part (d) diverges as the boundary is approached.

(c) Locked phase difference φ^*

The fixed-point condition is $f(\varphi^*) = \frac{\delta}{A}$. There are generally two solutions per period: one on each side of the triangle.

On the rising branch, $f(\varphi) = \varphi$, hence

$$\varphi_s^* = \frac{\delta}{A} = \frac{\Omega - \omega}{A}, \quad -\frac{\pi}{2} \leq \varphi_s^* \leq \frac{\pi}{2}.$$

To determine stability, differentiate the right-hand side of the phase equation. Away from the corners,

$$\frac{d}{d\varphi}(\delta - Af(\varphi)) = -Af'(\varphi).$$

On the rising branch $f' = 1$, so the derivative is $-A < 0$: this fixed point is stable. Thus the physically observed locked phase difference is

$$\varphi^* = \frac{\Omega - \omega}{A} \bmod 2\pi, \quad -\frac{\pi}{2} \leq \varphi^* \leq \frac{\pi}{2}.$$

For completeness, the falling branch gives

$$\varphi_u^* = \pi - \frac{\delta}{A} \bmod 2\pi.$$

There $f' = -1$, so the derivative is $+A > 0$; this second fixed point is unstable. At the two endpoint cases, the stable and unstable solutions meet at a corner.

(d) Drift period T_{drift}

Outside the entrainment range,

$$|\delta| > A\frac{\pi}{2},$$

so $\dot{\varphi}$ never vanishes. The phase difference moves through one full 2π cycle during each phase slip. First suppose $\delta > A\frac{\pi}{2}$, so φ increases. The elapsed time is

$$T_{\text{drift}} = \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{\delta - Af(\varphi)} d\varphi.$$

Split the integral at $\frac{\pi}{2}$ and insert the two linear formulas for f :

$$T_{\text{drift}} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\delta - A\varphi} d\varphi + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{\delta - A(\pi - \varphi)} d\varphi.$$

For the first piece,

$$I_1 = \left[-\frac{1}{A} \ln(\delta - A\varphi) \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{A} \ln \left(\frac{\delta + A\frac{\pi}{2}}{\delta - A\frac{\pi}{2}} \right).$$

In the second piece, set $u = \pi - \varphi$. Its limits become $\frac{\pi}{2}$ and $-\frac{\pi}{2}$, and reversing them gives exactly the same integral. Therefore $I_2 = I_1$, so

$$T_{\text{drift}} = \frac{2}{A} \ln \left(\frac{\delta + A\frac{\pi}{2}}{\delta - A\frac{\pi}{2}} \right), \quad \delta > A\frac{\pi}{2}.$$

If $\delta \leftarrow -A\frac{\pi}{2}$, the phase moves in the opposite direction. The time is still positive and, by the odd symmetry of f , is obtained by replacing δ with $|\delta|$. Consequently,

$$T_{\text{drift}} = \frac{2}{A} \ln \left(\frac{|\Omega - \omega| + A\frac{\pi}{2}}{|\Omega - \omega| - A\frac{\pi}{2}} \right),$$

valid for $|\Omega - \omega| > A\frac{\pi}{2}$.

As a check, $T_{\text{drift}} \rightarrow \infty$ when the entrainment boundary is approached from outside. For a very large mismatch, $\ln\left(\frac{D+a}{D-a}\right) \approx 2\frac{a}{D}$, with $D = |\delta|$ and $a = A\frac{\pi}{2}$; hence $T_{\text{drift}} \approx 2\frac{\pi}{D}$, the expected uncoupled result.